electronic brain guiding it seldom goofs off. It picks things up and puts them down with an accuracy of two thousandths of an inch! The Westclox Corporation of LaSalle, Illinois, uses the TransfeRobot to oil clock assemblies as they speed by on a conveyor belt. It oils eight precision bearings in a second. At the Underwood Corporation typewriter plant in Hartford, the robot picks up and inserts a small typewriter component into a close-fitting nest. The hand of the TransfeRobot can be adjusted to something more tender than a lover's caress, or it can grab things with the force of a vice. It can use mechanical fingers or electromagnets. For oozy stuff such as chocolate creams, it uses a gentle vacuum.

John Snyder, the maker of the TransfeRobot, says:

> *So far we have not been able to find any material or any shape or any size it can't handle. . . . We built an automated stamping line for Nissan Motors in Japan. It has six presses with TransfeRobot-type machines to take the stuff out of one press and feed it to the next. One man watches raw sheet metal go in and one more watches finished fenders come out. It replaces, oh, maybe 20 men altogether. But that's not all. You could extend the process right up to bolting the fender to a car. Even considering how cheap labor is in Japan, the thing saves money.*

History of Computers

Just as automated machinery replaces the tired muscles of man, computers are being developed that can replace the bored minds of men who are engaged in the repetitive production of goods and services in our industrial society. And just as automated machinery does a better job on repetitive tasks than the human hand, a computer outperforms a human mind. The circuits in an electronic computer can respond in less than a millionth of a second. This is over a thousand times faster than it takes the neurons in our brains to respond to incoming stimuli. Robert Theobald has pointed out, ". . . in the near future we will see that the computer can take over any structured task . . ."

Man worked toward the development of a computer for a long time. The simple abacus of ancient times was a first step. In 1671 Gottfried Leibnitz tried unsuccessfully to invent a mechanical calculating machine. "It is unworthy of excellent men," he wrote, "to lose hours like slaves in the labor of calculation." An Englishman named Charles Babbage worked out many of the principles of modern mechanical calculating machines in 1834, but he was unable to produce a successful model because the technology of his time could not turn out accurate gears. It was not until 1944 that the first true computer was produced by Howard Aiken, a professor at Harvard University. It is a sad reflection of our times that this pioneering computer was used to compute weapon trajectories for the U.S. Navy.

Aiken's first computer was soon overshadowed by the famous ENIAC, developed at the University of Pennsylvania. Though a great improvement, EN I AC had 18,000 vacuum tubes, was unreliable, and took too much space. Computers began to trim their waistlines and speed up their operations about 1958 when small reliable transistors and other solid-state components replaced the vacuum tubes. Computers today can multiply a half-million ten-digit numbers in a second. Many computers can make more calculations in an hour than an auditorium full of mathematicians could accomplish in their lifetime. In 1951 there were under 100 computers in operation in the United States. By 1965 the number had jumped

to 22,500 and is constantly accelerating. Some companies use up to 200 of them.

As long ago as 1959 the Sperry-Rand Corporation produced a computer that was able to handle 250,000 additions and subtractions of twelve-digit numbers per second. This enables it to make up a monthly payroll for 15,000 employees in only fifteen hours. Previously, it took 450 to 900 hours to do this job. Since only a part of the circuits of the computer are used while calculating a payroll, it can simultaneously solve scientific problems on the side!

By 1965 computers had been used in more than 700 specific tasks. At the time you read this page this number will have increased enormously. Major airlines use computers to give instant information on seats available on all flights. Stock Exchanges use computers to give instant stock quotations. Computers are now busy setting newspaper type, pawing over our income tax returns, controlling the flow of electricity of most power companies, helping you make long distance phone calls by ferreting out available circuits, navigating planes and ships, and providing railroads with instant information on where their freight cars are hiding out. Computers can land a plane in foggy weather without any human help. Our space ships that orbit the earth, spy on the moon, and whiz by the planets are highly dependent on computers from the design board to the final inch of their spectacular flights. Much of today's business, government, and science would be paralyzed if the computers were ever to demand a vacation.

It has been estimated that if there were no computers, the phone company would now have to hire all the working women in the country just to handle the flood of calls. Computers today control the production machinery in the petrochemical, petroleum, paper, and steel industries. At Western Electric's "Plant of Tomorrow" computers handle the billing, shipping, and warehousing; they order materials, write checks, and decide what to manufacture and how many. Time magazine has pointed out:

> *Computers have helped scientists to discover more than 100 new subatomic particles, and are busy analyzing strange radio signals from outer space. Biochemists have used the computer to delve into the hitherto unassailable secrets of the human cell, and hospitals have begun to use it to monitor the condition of patients. Computers now read electrocardiograms faster and more accurately than a jury of physicians. The Los Angeles police department plans to use computers to keep a collection of useful details about crimes and an electronic rogue's gallery of known criminals. And in a growing number of schools, computers have taken jobs as instructors in languages, history and mathematics.**
>
> *"Technology," Time, April 2, 1965, p. 86. Courtesy TIME; Copyright Time Inc., 1965.

Although recently invented, computers are rapidly transforming our civilization. "In a Chicago radio plant," according to Walter Buckingham:

> *1000 radios a day now are assembled by two men where two hundred had been required before automation. The duPont Company, using a computer at M.I.T., solved in thirty hours a chemical problem that would have required one man, working forty hours a week for twenty years to do the arithmetic alone.At the Institute for Advanced Study at Princeton an electronic computer works out weather predictions in three hours that would take one man with an adding machine three*

*centuries.In these last two cases, the job would not have been practical or economical to tackle without automation.**

** Walter Buckingham, Automation (New York: Harper & Row, 1963), pp.27-8*

A bakery is being automated so that grain delivered to the silo is not touched by human hands until the loaves of bread are ready for delivery. One bakery run by one man could supply the needs of an entire state. A local union with 1,300 members in 1959 had only 350 in 1963. Soon their plant will be automated further, and there will be only twenty-five workers, producing twice as much as before. Union leaders today are desperately worried about the livelihood of their members. Nevertheless, one union is automating its headquarters and reducing the staff from sixty to only six girls. When quizzed about this apparent contradiction, the union official explained that, "Business is business."

Cybernation

Cybernation has been described as the wedding of automated machines with computers. When you equip a factory with automated machinery that is controlled by a computer, you have taken the work out of production. There is little for people to do but turn the machinery on, step aside, and let it do the work.

For example, an automated cleaning fluid plant will have machines that mix and bottle the stuff. When cybernated, this plant will use a computer that is electrically connected to every machine, every storage bin, and every operating mechanism in the entire plant. The computer will have at all times full data on what is happening throughout the plant. It will digest this information and give instructions continuously to keep all parts of the plant operating at maximum efficiency. It will have a better grasp of what is happening second by second throughout the entire factory than any boss could possibly have. It never takes a coffee break or goes to the bathroom. The computer that controls this cybernated cleaning fluid plant will, among other things, send out orders for chemicals, bottles, labels, and other materials before they are needed. It will automatically shut down the plant or speed up the production depending on the need for the product. The computer will quickly spot any breakdown and order repairs instantly. It will maintain a continuous inventory.

Cybernation means that automated machines do all the work with a computer as the boss. The computer "boss" coordinates all activities in the plant so that no executives, secretaries, foremen, or other supervisory personnel are needed. A self-repairing, cybernated factory may operate 24 hours a day, 365 days per year without a single human. If a human were present, he would probably spend his time looking at dials and fighting boredom. The small crew that even today operates a modern cybernated oil refinery could do their work in dinner jackets and white gloves without soiling them!

"Ultimate automation based on atomic power," said Albert Einstein, "will make our modern industry as primitive and outdated as the stone age man looks to us today." It is possible to build an automobile plant

in which the raw materials are automatically put in one end and shiny automobiles run out another end, untouched by human hands. Cybernated systems that use almost no human labor can be developed to produce everything we use from the food we eat to the homes we live in.

Goods and Services Without Labor

Now, what does cybernation do to the shirt that we previously discussed which might take one hundred hours to produce without machines? If all raw materials are mined, raised, or gathered by cybernated machinery, and if shirts are produced in a cybernated factory without human beings, just how much labor is involved in the production of a shirt? It is conceivable that only five seconds of human time per shirt might be enough.

Further improvements might get this below one second per shirt. How much would a shirt be worth under these circumstances? Five cents? One cent? One-tenth of a cent? Would it be worthwhile even to worry about charging for a shirt if there were practically no human labor involved in its production or distribution?

Since any task done by human minds and human hands can theoretically be cybernated on a repetitive basis, the advance of modern technology will almost eliminate the human labor cost of services. Services such as dry cleaning are now being automated. In the future haircuts, manicures, beauty parlor services, laundry, and the servicing of automobiles will be performed on a cybernated basis. The ability of cybernated machines has been described by Donald N. Michael:

> *Cybernated systems perform with a precision and a rapidity unmatched in humans. They also perform in ways that would be impractical or impossible for humans to duplicate. They can be built to detect and correct errors in their own performance and to indicate to men which of their components are producing the error. They can make judgments on the basis of instructions programmed into them. They can remember and search their memories for appropriate data, which either has been programmed into them along with their instructions or has been acquired in the process of manipulating new data.Thus, they can learn on the basis of past experience with their environment. They can receive information in more codes and sensory modes than men can. They are beginning to perceive and to recognize.**
>
> **Donald W. Michael, "Cybernation: The Silent Conquest" (Santa Barbara, California: Center for the Study of Democratic Institutions, 1962), p. 6.*

Humans will not even be required to maintain the factories of the future. Cybernated factories will be designed to operate for many decades without repair by men. Routine maintenance and repairs will be performed by machines. Improved metals and designs of the future will make machines almost impervious to wear and tear. Even today Western Electric makes complex switches that work so

flawlessly that a single failure in five million operations is considered below par. Some relays now in use will perform a billion switching operations in their lifetime.

The Human Use of Human Beings

Dr. Norbert Wiener, the "Father of Cybernetics," wrote:

> *It is a degradation to a human being to chain him to an oar and use him as a source of power; but it is almost an equal degradation to assign him purely repetitive tasks in a factory, which demand less than a millionth of his brainpower.*

The advent of cybernation may be regarded as an emancipation proclamation for mankind. Its thorough application will at least enable man to have the highest conceivable standards of living with practically no labor. It will free him for the first time from a highly structured and outwardly imposed routine of repetitive day-by-day activity. It will permit him to return to the Greek concept of leisure where all work was done by slaves and men had time to cultivate their minds. In the future each of us will command a million slaves. These will be mechanical and electrical slaves, not the degrading use of a human being to do the work so that another may live an abundant life.

Computers are today in a very early stage comparable to the Model T Ford. At the present time the largest electronic computers have only about 1/10,000 of the associative powers of human beings.

Computers today are generally programmed to operate in specific ways. We are only beginning to design them to observe a large range of outside data and creatively handle this information.

Although computers are just out of their teen years, some of them are already beginning to show a potential for originality.

"The present level of these learning machines," said Dr. Norbert Wiener :

> *is that they play a fair amateur game at chess but that in checkers they can show a marked superiority to the player who he programmed them after from 10 to 20 playing* hours of working and indoctrination*. They thus most definitely escape from the completely effective control of the man who has made them. Rigid as the repertory of factors may be which they are in a position to take into consideration, they do unquestionably --and so say those who have played with them-- snow originality, not merely in their tactics, which may be quite unforeseen, but even in the detailed weighting of their strategy.*
>
> ** Norbert Wiener, "Some Moral and Technical Consequences of Automation" Science, Vol. 131, No. 3410, May 6, 1960, p. 1306 . Copyright 1960* by the American Association for the Advancement of Science.

The micro-miniaturization of computer components may enable man to build computers in the future that will have a thousand times more associative power than any human brain. Computers will be designed that will have sensory receptors in all parts of the world which will give them immediate information on anything significant that occurs. A master computer in the future will be able to gather, digest, and analyze all recorded facts and information-a fantastic task that is impossible for any human being. The range of facts and formulas .s so extensive today that it is often impossible for a scientist to keep up with new developments even in his specialty.

Only a computer will be able to handle the integration of all knowledge and come up with decisions that will be based on the full range of relevant data. In the future computers will not only be able to think as well as men, they will be able to exceed man enormously in the capacity to digest facts and information. They will analyze the data and come up with solutions to problems that will enable man to obtain what he wants on earth. It would be impossible to achieve the value system discussed in Chapter 4 without the thorough use of automated machinery and computers integrated into a world-wide cybernated complex.

Cybernation can transform our entire world into a Garden of Eden The goods and services that we desire will be available without repetitive human toil. And the Tree of Knowledge will bloom for the fulfillment of everyone. The biggest portion of the prime years of one's life will no longer be structured by the need for a weekly pay check. With cybernation household jobs will no longer saddle women with boring day-to-day routines.

Men and women will for the first time in their lives be free to inquire into their own needs, to face themselves, and to work out satisfying patterns of life based on their own feelings and thoughts. *"The liberation of people from tasks unworthy of human capacity,"* said Gerald Piel, *"should free that capacity for a host of activities now neglected in our civilization. . . ."**

* Gerald Piel, "Consumers of Abundance" (Santa Barbara, California: Center for the Study of Democratic Institutions, 1961), p. 9.

The rat-race will be over. Society will require relatively little from any of its members. John F. Kennedy's famous dictum, "Ask not what your country can do for you; ask what you can do for your country," will be reversed. People may for the first time enjoy an abundant life made possible by the creative intelligence of man. With a cybernated technology it will be possible for all humans to live better than if each person were to have a million slaves at his command. Even a millionaire today has a crude, harried life compared with the smooth, fulfilling pattern that all people may achieve in a cybernated society.

We have asked how you would like a guaranteed income of $100,000 per year. In the 1960's in the United States, the average family has had income of around $6,000 per year. However, when there is almost no human labor cost to producing cybernated goods and services, it will be possible for everyone to have almost anything in any quantity! One hundred thousand dollars per year or $1,000,000 per year—it won't matter. Of course, with practically no labor cost and limitless nuclear energy, things won't have price tags. Prices are only a way of distributing the loot when there is not enough for everyone.

Man's future will be a thousand times more exciting than his past. For the first time man will be free to work out any system of values that he desires—and to achieve these values. For the first time man will be in a position to make a thorough application of the scientific method of thinking so that his knowledge has

predictability—so that he can solve his problems and not just patch them up crazy-quilt fashion. For the first time it will be possible to have a cornucopia of goods and services that will not be obtained by the sweat of someone's brow.

"This is a time of transition. . . ," said Adlai Stevenson, "from the ancient problem of sharing scarcity to the modern problem of distributing abundance." Cybernated technology will for the first time permit us to realize our human potential. We may achieve the deepest measure of life; we may enjoy the highest level of liberty; and we may have maximum scope in our pursuit of happiness.

7.Away We Go!

In the next section we will leap into the twenty-first century. In previous chapters we have found that man escaped from the jungle a relatively short time ago. As might he expected he brought with him many primitive habits of thinking and feeling that still plague us today. We have discussed in detail the triple foundation on which we are making a projection of our twenty-first-century civilization. If you can accept our value system, if you feel that the scientific method of thinking will play a dominant role in the future, and if you understand the impact of a cybernated technology that will produce goods and services with practically no human labor, then we feel that you will find our projection thought provoking.

We do not believe it should necessarily take one hundred years to accomplish the technological and sociological changes that will provide mankind with this cybernated Garden of Eden. Some of the things we are anticipating may be well under way by the time this book is published. If the improvement of our society were given the same priority that the development of the atomic bomb was given during the last war, it would be possible to achieve most of the features we project for the twenty-first century in time for us to enjoy them during our lifetime. U Thant, Secretary General of the United Nations, said:

> *The truth, the central stupendous truth, about developed countries today is that they can have—in anything but the shortest run—the kind and scale of resources they decide to have. ... It is no longer resources that limit decisions. It is the decision that makes the resources. This is the fundamental revolutionary change—perhaps the most revolutionary man- kind has ever known.*

Books and articles describing the future usually deal with space ships and other technological marvels and gadgets. They wisely stay away from upsetting the reader's values or challenging the age-old patterns by which he lives. People are not usu-ally threatened by technological change but they get emotional when someone proposes a social change. A twenty-year-old farm boy will join the Air Force and fly jet planes faster than the speed of sound. His grandparents below will hop into their 350 horsepower chrome-plated monster and speed over the expressways. But it took over a century for a good part of the people in the United States to recognize that we should respond to human beings as individuals, instead of on a basis of race, creed, or color. And that battle, unfortunately, is still going on.

As we leap into the future, we must make every attempt to avoid being bogged down by tradition and the

"wisdom" of the past. It is useless to try to fight change. It is much more fruitful for human beings to control and channel intelligently the accelerating flow of events. Over a half-century ago the United States Electric Light Company gave its dynamic inventor Hiram Maxim a $20,000 annual life pension and exiled him to England. They felt they needed to get rid of him because his brilliant mind kept inventing improvements. His creative ability made their equipment obsolete before they had time to pay for it. Maxim produced some of his greatest inventions in England. At the time that he was being knighted for his outstanding accomplishments, the United States Electric Light Company was going out of business.

Probably the only thing we can know for sure about the future is that it will be very different from what we have today. But whatever difficulty we may have in trying to understand life in times to come, it is little compared with the trouble people in the twenty-first century will have understanding the way we do things today. In the future people will find it almost impossible to believe that human beings could have organized themselves into nations and then could have used scientifically designed weapons to slaughter each other. As they watch movies of the past, they will be astounded by the tobacco smoke emanating from the nostrils and by the ostentatious clothing and dangling jewelry. They will find animal emotions of hostility and jealousy most incredible. Individuals in the twenty-first century will not look back with nostalgia to a world threatened by atomic oblivion, with economic and political activities vitiated by greed and hypocrisy, and with mental disorders growing at a phenomenal rate. How crude and pathetic we will appear in the eyes of our descendants !

Civilization Lightens Man's Burden

One measure of the degree of civilization at any given time is the extent to which it requires individuals to sacrifice themselves. In the past millions of men were required to sacrifice their lives during the recurrent wars. These individuals usually sacrificed themselves willingly, for they had been conditioned this way.

An Englishman once described America as, "A place where everybody furiously works overtime making labor-saving machinery." To operate the economy of industrial nations today, many individuals are required to sacrifice the prime portion of their lives in an eight-hour-per-day pattern of labor. They are conditioned so that they do not usually consider it a sacrifice. The work week is getting shorter. Some people in the preceding century had to work twice as many hours to make a poorer living.

Furthermore, working conditions are improving, fringe benefits increasing, time-and-a-half more common —if not double time —vacations longer, and every now and then someone manages to convince himself that some part of his job is interesting. Men and women are working to get money to buy things they want, or they are working for recognition in terms of titles and achievements. But the fact remains that the prime portion of the lives of most twentieth-century men and many women is consumed by more or less compulsory, more or less monotonous, more or less repetitive, more or less boring tasks which are associated with a pay check.

The mature society of the future will burden man with a minimum of obligations. Most societies of the past and present could not operate unless its citizens were heavily committed and obligated to certain set

patterns. But as we shall see, the automated world of the future will for the first time free mankind from these heavy obligations to the group. He will be able to face himself deeply and fundamentally. No longer will his parents, his boss, and his country tell him how he should act. Our future society will require minimal work, criticism will not be considered disloyalty, and diversity in sexual and family patterns will be possible.

In the future individuals will do most for their social group by developing themselves into dynamic, happy human beings. Men and women will ask themselves: "What fills me as a human being? What things add to my feeling of worth? What do I enjoy? What do I really need? What things make me feel intellectually vital, emotionally warm, and physically strong? What makes me feel ten feet tall, that life is glorious, and that today is wonderful?"

Steps to the Future

The rate at which we progress toward the better world of the future will depend upon how rapidly we use the scientific method to test out various solutions to our problems. In the past we have fired professors who advocated doing things that were different from the present mores of our tribe. In the future we must take these creative men and give them the facilities they need to test their ideas scientifically.

We must plan to increase the available power and energy in all parts of the world to enormously higher levels. Intelligence guided by scientific methodology must be applied to the technological and sociological reconstruction of our entire planet. A cybernated food production system must be designed to meet the needs of a stable world population. Areas must be set aside for industry that will be coordinated with a vast international transportation system. A product that is cybernetically manufactured anywhere on earth should be cybernetically delivered to almost any building on earth in less than twenty-four hours. We must open our eyes and minds to use to the fullest man's enormous ability to create. Knowledge is exploding. It is reported in Schools for the Sixties, a volume sponsored by the National Education Association that over the last 2,000 years, knowledge doubled for the first time by 1750, for the second time by 1900—(150 years later) for the third time by 1950—(50 years later), and for the fourth time by 1960— (10 years later).

The world's supply of technical knowledge is now doubling every seven years. Most of the scientists who have ever lived are alive today. "By now," said Dr. A. C. Hall, the Defense Department's Deputy Director of Research and Engineering for Space, "we seldom doubt the technical feasibility of anything."

Dr. George Gallup in his book The Miracle Ahead has shown us the type of thinking that will enable man to achieve a maximum of "Life, Liberty and the Pursuit of Happiness." Dr. Gallup asks:

> *Can man perform the miracle of lifting himself to a higher level of civilization?*
> *The answer is Yes—unequivocally. Man is clearly in charge of his own evolution; he can proceed at a pace that he himself sets.*
> *He can solve any problem that comes within his purview—even the problem of war. The great advances made in physical science can be paralleled in social science.*

Man now has the procedures for dealing with the problems arising out of his social existence—problems that the methods of physical science can not adequately explore or illuminate.
Man has scarcely begun to make use of his almost limitless brain power, either individually or collectively. Lack of progress in dealing with the affairs of mankind can be traced to a simple truth: man has never made a concerted and persistent effort to solve his social and political problems. His inventive genius has been confined almost exclusively to the production of better tools and instruments.
The next great move forward can now be taken. All that is required is a firm belief in man's great potentialities and a readiness to accept change.
*Man is still young on the face of the earth; civilization is still in its infancy. Homo sapiens has not yet realized his strength and his greatness; nor does he see, except dimly, the heights to which civilization can reach.**

* George Gallup, The Miracle Ahead (New York, Evanston and London: Harper & Row, 1964), p. 203.

Mankind today is in a period of challenge and opportunity. Fascinating, unexplored territory lies before us. The explorations of Columbus will seem like child's play in comparison. By joining the forces of science and technology throughout the world in a common endeavor, we can eliminate man's inhumanity to man. We can reconstruct the whole of the world environment to give every person on earth what he needs to live a fulfilled, abundant life. We can build a new society with sufficient flexibility to correct its own errors and to meet any challenge that lies ahead. In the future no individual will ever stand alone. The unlimited horizons of the humanistic-scientific-cybernated future will be the most exciting adventure in the history of mankind.

PART II
A Projection of Our Future

8. At Home in the Twenty-first Century

Scott and Hella have been asleep two hours. They will probably awaken in about an hour. In the previous century it was considered normal for people to waste about one-third of their lives sleeping. One way to attack the problem of increasing the effective life-span has been to make two or three hours of sleep as effective as eight or nine hours previously was. Several genetic improvements, an increase in oxygen in

the sleeping chamber, plus the development of deeply relaxed personalities almost free of hostility and tension has proved successful.

Further genetic improvements are expected to minimize even more the amount of sleep required. The reduction, if not elimination, of sleep is not desired for its own sake. This world of the latter part of the twenty-first century is stimulating and challenging. There is so much to do and see. The limitless intellectual horizons, the esthetic delights, and the sensual feelings are too many and varied to be fully savored in the average lifetime of 200 years.

Although Scott and Hella are asleep, they are surrounded by great dynamism. Everything around them is being controlled by their home computer, known as the cybernator. This small computer is built into the wall and they never see it although it is in use every minute of the day. They have gradually trained their cybernator to meet their needs in thousands of ways. Through the cybernator Scott and Hella can verbally command any mechanism in their apartment. The cybernator also handles their messages to the Correlation Center.

The automated bed upon which Hella lies nude and unrestricted by clothing, sheets, or blankets, responds in a living way to support her body. There are no pressure points, no creases. This soft membrane gradually moves in a rhythm and pattern that over the years has proved to be most relaxing to Hella. The rhythm that is most relaxing to Scott during sleep is slightly different and the cybernator is also attending maximally to his needs. Since people change, the input sensors of the cybernator in the apartment are constantly scanning to explore whether they would have more relaxation if the sleeping membrane were to undulate in a different manner.

If Scott and Hella were to have a thousand servants, they would not receive the services that are available to them through the cybernator. Not only are the undulations of the bed constantly adapted to meet their needs, but other aspects of their sleeping chamber are being maintained by the cybernator to give maximum restfulness and comfort. A controlled, ionized atmosphere adds to their feeling of well-being. The temperature, humidity, mixture of air, and background music are constantly being adjusted to meet their individual needs at the moment. If one of Hella's feet drops three degrees below her optimal temperature range, an infra-red beam immediately brings it back to the desired level.

Sleep in the twenty-first century is no longer a haven from the trials and tribulations of the day—"the balm of tired minds". Scott and Hella have never experienced nightmares—these are part of the long list of things about previous centuries that they are really unable to understand except on an abstract level. In the past centuries, conflicts produced tension-filled lives that were temporarily relieved by alcohol and tranquilizers. The repressions and injustices of the day were expressed at night as disturbing dreams and nightmares. In Scott and Hella's world one's feelings and impulses are accepted both in thought and action so that hours of sleep are not vitiated by the romping of repressed feelings. Scott and Hella rarely dream, but when they do, their dreams usually revolve around pleasant things they plan for the next day.

Scott and Hella's lives are not governed by rigid schedules. There are practically no deadlines, and there is no need for them to awake at any particular time. The sensing extensions of the cybernator are able to determine when their bodies have absorbed all the rest they needed. Gradual changes are made that will prepare them to awaken with a full-of-energy feeling. The temperature of the sleeping chamber is reduced several degrees. The lighting of the chamber is increased and background music of a type enjoyed by

Scott and Hella in the morning will soon begin.

As Scott and Hella awaken, they have a feeling of anticipation. A new day is here. New and interesting thoughts with new beauties to be experienced and new sensations are to be felt! Lying together for the first few minutes of the day, Scott and Hella chat warmly about their feelings and plans. They decide to talk to some friends who live 10,000 miles away. They give verbal commands to the cybernator which makes a connection with the Correlation Center that is immediately relayed by satellite to their friends. They are able to exchange experiences and thoughts via a three-dimensional color transmission. They feel as close as if they were together in the same room.

The Cybernated Hygiene Area

With kitten-like playfulness Hella slaps Scott on the buttocks and runs toward a cylindrical chamber. Scott catches her just as she reaches the entrance to the twenty-first-century bathroom.
Although she is forty years old, Hella looks like a young girl of eighteen. She has light brown hair with sparkling brown eyes. Her mouth is expressive and perfectly spaced teeth show when she laughs. Her breasts are firm and slightly smaller than the average of previous centuries. Her hips are beautifully formed but not wide. Her buttocks and thighs are softly rounded. Laughingly, Scott pulls her inside the cylindrical walls of the shower. Air and water, mixed under pressure to form a soft, cleansing spray, delightfully comb every part of their bodies at a speed and pressure that they have taught the cybernator. No soap is used; ultrasonically activated water loosens any clinging particles. This bath not only cleans their bodies but also furnishes the most delightful tactile sensations with sprays that tingle and massage every part of them.

Scott is about the same height as Hella. His handsome face responds to the stimulating shower. Like Hella, Scott does not look his forty-five years. He begins to sing in a resonant voice. Hella, in self-defense, harmonizes with him with a vibrant tone. Scott and Hella have almost equally strong muscles. They move gracefully and gently in a way that suggests hidden strength. They are dried in three seconds as they walk through the air wall that acts as a shower door. This air wall is a high-speed sheet of warm air jetted from the top, bottom, and sides of the shower opening. While they are in the shower, the sleeping chamber automatically cleans and sterilizes itself.

Scott and Hella then lean their heads backward to fit into a niche designed to groom their hair. In forty-three seconds this cybernated beauty shop trims the hair and arranges it in any mode that Scott and Hella select. Over the years they have trained the cybernator in the patterns that they prefer for their hair. The job of grooming is performed by an electronic complex that sets the hair by beamed positive and negative electrostatic charges. A one-second, modified laser emission gives a permanent set that remains until a different style is desired. Hella seldom uses rouge, eye shadow, eyebrow pencil, or other artificial techniques of past centuries. The people in the new civilization are no longer physiologically and psychologically exhausted by insoluble problems, emaciating responsibilities, atmospheric pollution, and poor nutrition. Neither men nor women care for artificial decoration. They feel beautiful and attractive in themselves. Their loveliness comes from inside—no outside veneer is needed.

At this time shaving has disappeared. Hair no longer grows in areas where humans do not want it. No mouth washes or irritating chemical agents are used that might affect the living tissues of the mouth. Cavities are unknown, for there has been an increase in the hardness of the enamel, and foods are designed to inhibit decay of the teeth and disease of the gums. Because of the high degree of intestinal health, bad breath is almost unknown.

The lavatory and toilet are recessed in a corner of the bathroom area. The water turns on whenever the hands are placed above the lavatory. In toileting, the individual sits upon a soft ring covering a soundproof bowl. During elimination or defecation the waste matter, together with all odors, is drawn into an opening. Instead of towels or tissues there are water sprays which automatically clean the rectal and pubic areas, and warm air rapidly dries them. This natural function is no longer unattractive as in previous centuries.

A Medical Checkup

Scott and Hella step into a cabinet and automatically trigger a ten-second series of tests that help them achieve the highest level of health. The mechanism records their weight so that the cybernator can check gain or loss. If an upward trend in weight appears over a period of time, the cybernator orders the food production mechanism to decrease the calories without making any noticeable change in the bulk or taste of the food. The cabinet also measures blood pressure differentials throughout the body. It produces an electrocardiogram and instantaneously compares it with previous electrocardiograms. The blood in capillaries on the retina of the eye is given a spectro-analysis. The heartbeat, respiratory rate, brain-wave activity, and many other measurements are made and instantly compared with long-term norms for the individual. The cybernator does not over-respond to individual readings of any specific day but, instead, analyzes them for physiological trends. The Correlation Center compares them with norms based on over two billion people.

Almost everything that happens in the human body is accompanied by electrical and chemical changes that can be picked up and recorded by the advanced medical engineering of the twenty-first century. The casual ten seconds spent in this medical cabinet give Scott and Hella a daily checkup that may add years to their lives. All colds, viruses, and infectious diseases have been eliminated for many years. Only long-term deterioration of body organs is still a problem.

When necessary, conditions in their environment are altered to help them maintain optimal health. Many of the environmental changes, such as the adjustment of calories to preserve optimal weight, are performed automatically. Every effort is being made by medical technicians to automate all conditions involving health so that no conscious control is necessary to achieve the highest level of energy and longevity.

Scott and Hella walk into the dressing area and hold out their arms. Their garments release themselves from hangers and adhere to their bodies. Their measurements are stored in the Correlation Center. Whenever they need a garment, the cybernated machines tailor a unique one for them in the style and material they select.

The clothing is extremely thin and soft, yet it has great strength and flexibility. The material lives, breathes, and reflects— or absorbs—light and heat as needed to keep the body temperature even. Through energy generated from light, the material obtains electrical potential required to operate the garment's electromechanical responses. It has no bulging surfaces or tension-producing areas of stress. In any scuff contact a flow of electro-migratory materials maintains the garment in constant repair.

The material can assume any color or become transparent. The clothes clean themselves and usually do not need attention during their expected ten-year life.

Delicate shoes that flex coordinately with the movement of the musculature of the foot are entirely free of local pressures or undesirable friction. They breathe as the wearer walks about.

The feet are maintained at the most comfortable temperature, regardless of the weather. A membranous material will migrate to any part of the shoe to enhance movement, comfort, durability, and wear. They are also self-cleaning.

Snack on the Balcony

Scott and Hella often eat on the balcony that overlooks the naturally lovely wooded areas of the twenty-first-century cities. Over 83 per cent of the land in the cities is maintained as park and recreation areas. For the past forty-two years all the cities have had full weather control. The Correlation Center arranges for the degree of variation from season to season that people find most pleasant. Rain, snow, and windstorms no longer inconvenience people in urban communities.

Scott and Hella live in a circular, multi-story apartment building that is over a mile in diameter. It contains 300,000 living-units designed to meet human needs in every possible way. All walls, doors, and windows are soundproof. The nearest building they can see from their balcony is about a half-mile away. Few suburban homes of the past, even if built on forty acres of land, offered the absence of distractions they are able to enjoy.

As they make themselves comfortable on the balcony, they give vocal commands to the cybernator which produces the food they have ordered in five seconds. As a part of their training to live in the twenty-first-century world, Scott and Hella have become familiar with 325 selections of food. These selections seem to encompass fairly well the entire range of taste, smell, and texture combinations that most people enjoy. It has been found that three basic food mixtures can be electronically altered in one second's time to give variations in color, taste, and texture that make up any desired menu.

In an experiment several years ago some twentieth-century foods, as prepared by gourmet restaurants in various cities of the world were offered to a group of twenty-first-century people. They found the twentieth-century food lacking in taste depth, nutritionally insufficient, and actually harmful in some ways.

Scott and Hella do not use the primitive knife, fork, and spoon of former times. Their exotic foods are picked up by a glass-like rod that is charged electrostatically so that portions of food adhere to it. They do

not have to penetrate the food; they simply bring the glass rod near it. There is no dripping or dropping of food. By varying the charge at the tip, even liquids adhere. When they complete their meal, the implements and dishes are lowered into the table, where they are automatically cleaned and sterilized.

The Dynamic Living Area

After breakfast Scott and Hella go to the largest room in their apartment. Half of this room is devoted to a teleprojection area large enough for life-sized, three-dimensional figures. An index scanner enables Scott and Hella to select whatever they enjoy most—concerts, plays, current events, informative subjects. The world's forests are no longer being chewed up to make newspapers. All past or present news is available on the telescreen, and reusable electrostatic copies can be made if desired. The cybernator in their apartment has already made a list of the programs of the day. It has put a red dot beside the type of program that Scott prefers and a yellow dot beside the type that Hella usually selects.

Scott and Hella nestle in a contour chair as they activate the three-dimensional teleprojection area. This contour chair acts as a living support that produces a comfort previously unknown. If Hella raises her arm, the chair will extend itself upon command to support the arm in any position desired. Whenever the legs are moved, it grows to support them with balloon-like softness. It gives them a physical freedom when sitting or reclining. The furniture of previous times tended to force people to sit in predetermined ways. In the new world of the twenty-first century, the individual has the freedom to select.

Most of the mechanisms that free Scott and Hella from drudgery and permit them to be served as guests in their own home operate automatically. Push buttons, dials, and levers are seldom used. Almost every machine in the new world is voice-actuated and responds instantly to spoken orders.

The interior of the living room has a large, dome-shaped ceiling with soft, colorful lighting flowing without any visible source from all portions of the wall. All of the electronic mechanisms that control the interior of their apartment are built into the walls. There are no gadgets bulging out here and there. The eye meets only pleasing contours of an organically designed interior. The walls of the twenty-first-century living-units are capable of infinite variety. At times they appear transparent. At other times they seem opaque. Often they reflect color or combinations of color in pleasing blends and designs. Teleprojected pictures, sculpture, and flowers are tastefully distributed through the apartment. They automatically change each day. There are no locks on any doors. In a world of abundance and sanity, they have no function. Almost everything in the apartment is fireproof and free from deterioration and will remain so for the long life of the building.

The entire apartment is maintained and cleaned continually by automatic mechanisms silently operated by the cybernator. No brooms, vacuum cleaners, or other manual paraphernalia are needed. It is almost completely dust-free. All surfaces are gracefully contoured so that there are no cracks or corners to permit dust to gather. Most surfaces have an electrostatic charge that repels dust and keeps it floating in the air to be filtered out. Since the air pressure in the apartment is slightly higher than outside, no dust flows in.

All materials and mechanisms in the living area are designed to last over one hundred years. Scott and

Hella probably cannot recall any inconvenience due to a mechanical failure. The outside walls and roof of the apartment building are made of ceramic-like materials which require no painting or maintenance and have a life expectancy in excess of 500 years.

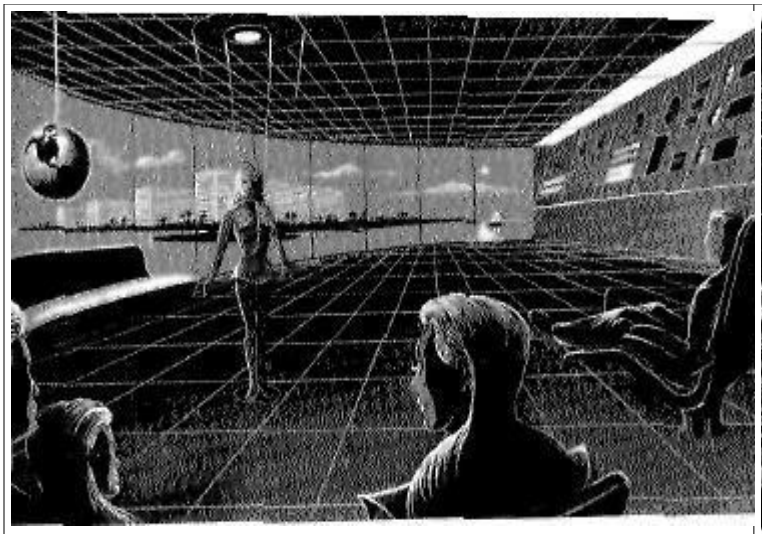
The versatile living areas are the focal points of the intelligence of the world environment. They are connected electronically with the Correlation Center, which, in turn, is connected with practically everything on the planet. When Scott wishes, he can contact any region in the world. He can talk with almost anyone in the world at any time. It is possible for him to attend any conference, to observe almost everything going on in the world in three-dimensional, color teleprojection without leaving his apartment.

Not only does their apartment enjoy two-way communication throughout the world, but it is also connected to receive anything they want directly from any part of the globe. Stores and shopping centers are regarded as inconvenient folkways of pre-twenty-first-century civilizations. When Scott and Hella want any personal item or any apparatus, they need only order their cybernator to produce three-dimensional models for their selection. Sometimes a basic model may have hundreds of optional attachments. This gives them an opportunity to order a customized version that meet their needs exactly.

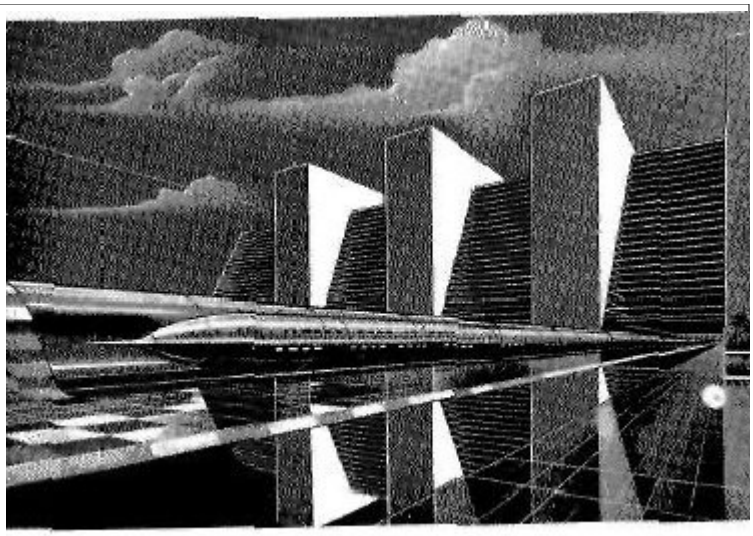
When they select what they want, their cybernator immediately communicates this to the Correlation Center. In less than a second the order is registered at the nearest industrial complex. Within minutes this item is fabricated, packaged, and sent on its way in a high-speed system of tubes twenty feet in diameter. This high-speed package is electronically guided by the symbols representing Scott and Hella's address. Their package travels at the rate of 250 miles per hour until it arrives at their apartment.

During the entire process not one human hand or brain has been involved in filling their order. It is possible that the object they have ordered is unique in the entire world, for perhaps no one else has ordered that particular combination.

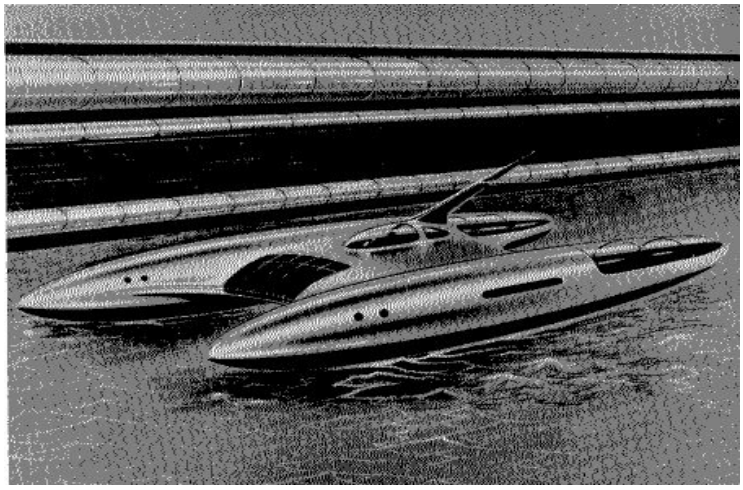
There is no scarcity of anything. Scott and Hella are free to order as much as they wish, for no human lives are consumed in meeting their needs. Whatever they want results only in a momentary blip in the cybernated machine complex of the twenty-first century.



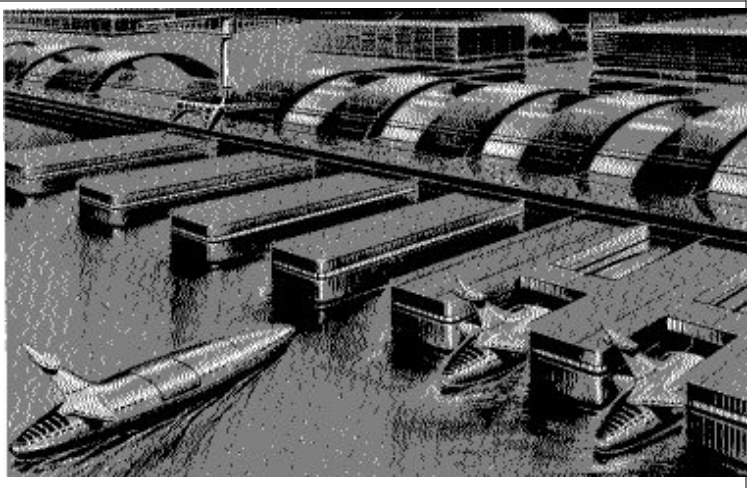
Living room *of Scott and Hella's apartment*
The group is watching a three-dimensional teleprojection. The panel on the right contains facilities for shopping and rapid delivery, delicious instant food. sending and receiving information, etc.



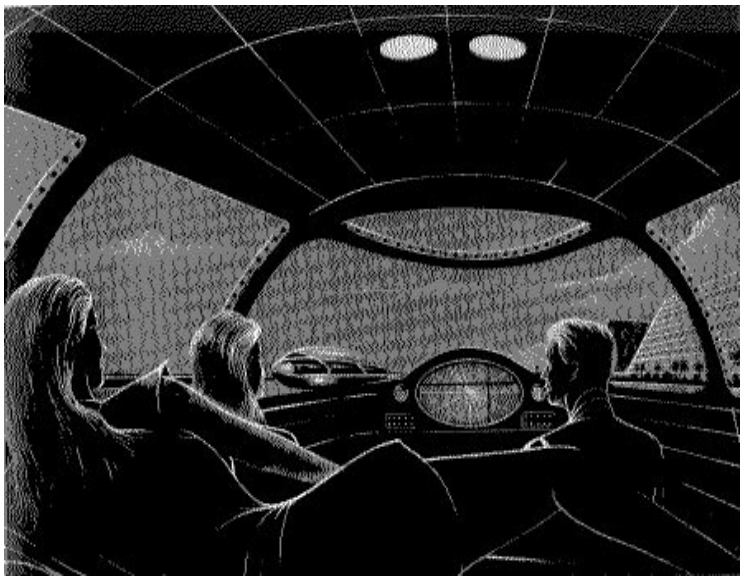
Linear-acceleration train
This train is magnetically propelled on a cushion of air at 2,000 miles per hour. It has no engine or wheels. The electrostatically charged point reduces air resistance. Its safe cybernated operation requires no personnel.



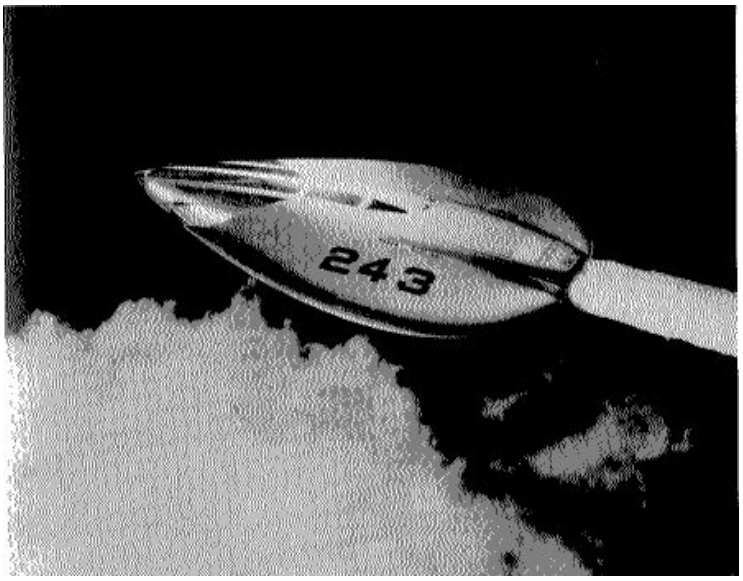
Floating research vessel
Children as young as five years participate in interesting research projects such as this oceanographic research vessel's analysis of currents near the Bering Strait dam. Trains connecting the continents travel through the upper tube.



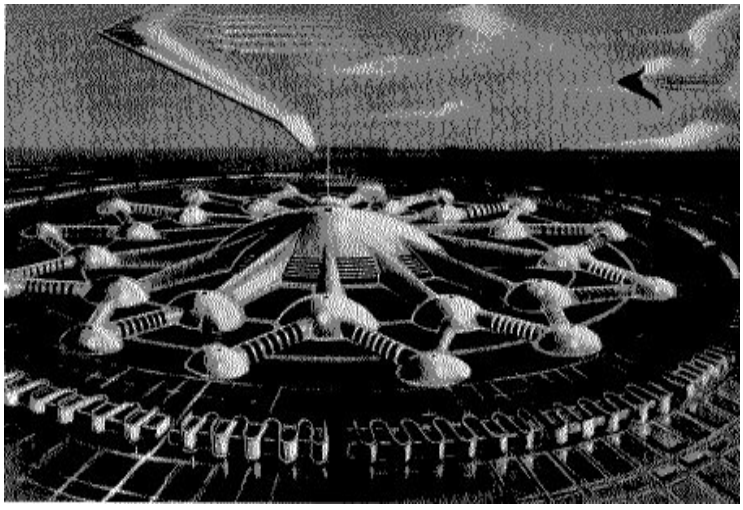
Cybernated freighters
These rust-free titanium vessels load and unload themselves and navigate to any port without captain, crew, or dock workers They carry enough atomic fuel to power the ship for fifty years.



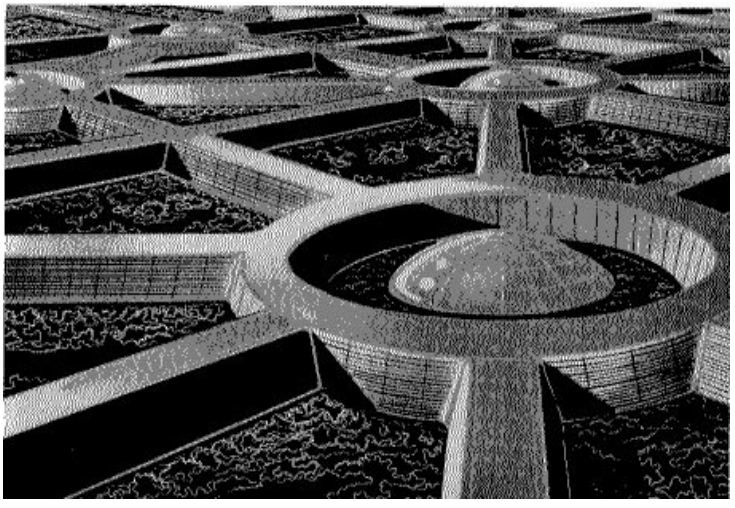
A twenty-first-century car
These wheelless cars glide quietly and safely while people relax You state where you want to go, and they guide themselves without attention. They are refueled every five years and usually operate twenty years without repairs.



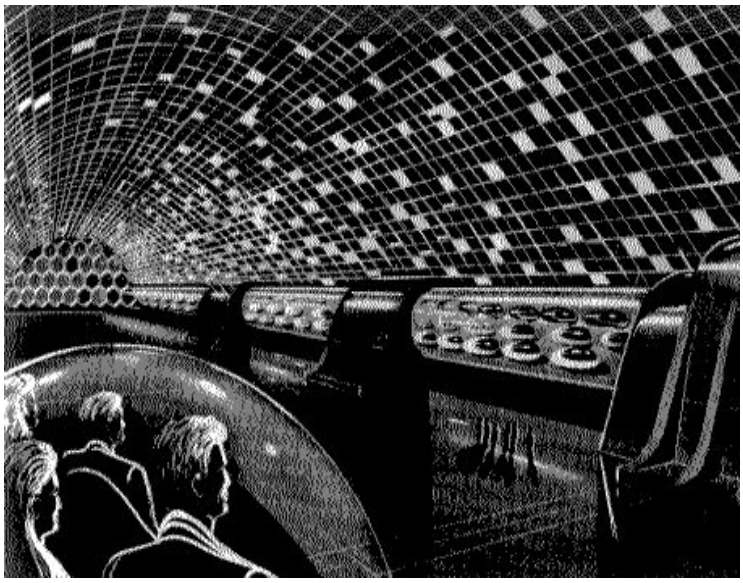
Ion-propelled aircraft
This pilotless aircraft picks up Scott and Hella from the top of their apartment and cybernetically transports them to the Exuma Islands It is equipped with the comforts and conveniences of the twenty-first-century apartments.



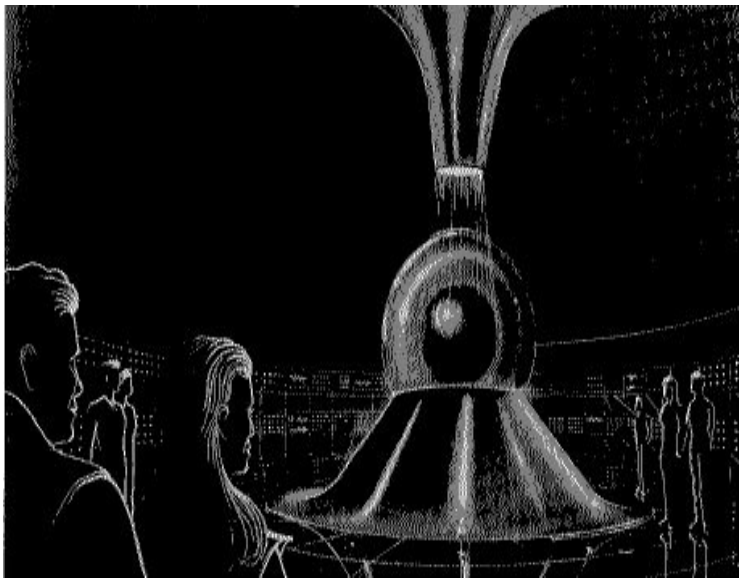
Power and research center
The nuclear power plant that Hella visits lies 300 feet below the central dome that contains the computer complex. Domed laboratories branch off from the center and are conveniently interconnected.



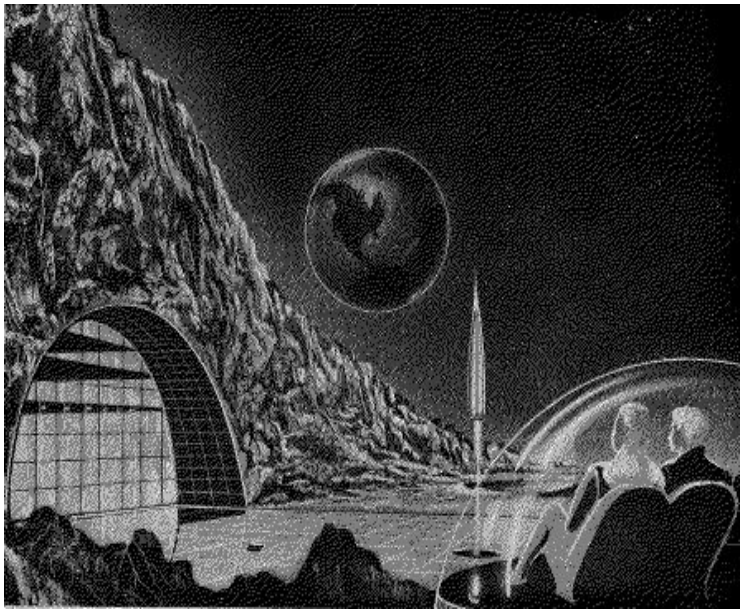
City design
Each of these spaciously designed fifteen-story living complexes overlooks natural areas of forests and lakes that are a minimum of one-half mile wide by one mile long. Transportation is handled in underground raceways.



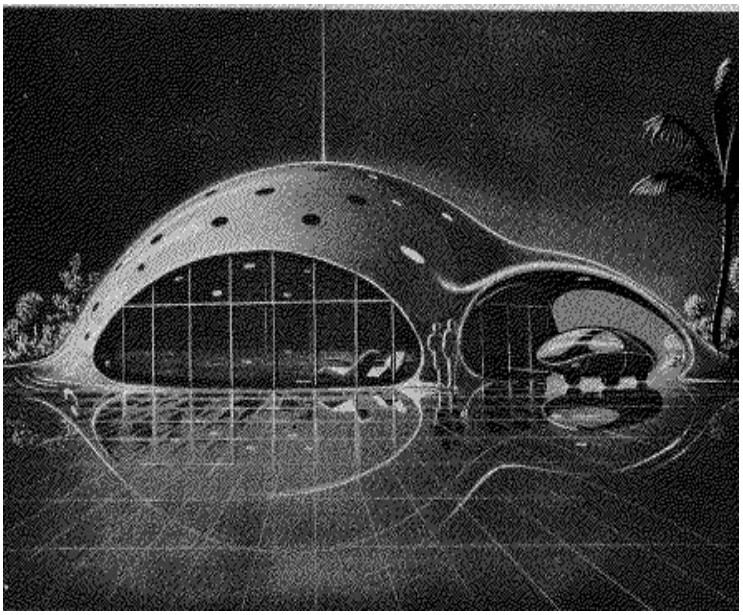
Industrial production
A continuous stream of levicars is silently electroformed in one of the 1,000-foot flow lines. All objects are formed molecularly as a whole instead of being fabricated in parts and then assembled as in the pre-cybernated age.



Corcen
This six-foot sphere which serves as the world correlation and knowledge bank has trillions of inputs from all over the globe that enable it to serve every individual and coordinate the humanized man-machine symbiosis.



From the Moon
Scott and Bella's observation dome overlooks the laser-cut excavation in the rock which protectively



Suburban home
This home (including terrace) is electroformed in one piece and set in place by a cargo levicraft.

encloses the multi-level interior environment. In place of exterior doors, there is a system of translucent thermal barriers.	Although cybernetically produced, the limitless choice of design elements and equipment make it uniquely individual.

Fulfilling Interests

Scott has strong interest in medical engineering. A writer in the twentieth century would have said that medicine was Scott's "profession". In the new world, this terminology is not appropriate since the primitive system of jobs, wages, fees, and money has been outgrown. Scott finds the human body and its myriad mechanisms particularly fascinating. He enjoys playing a part in experiments designed to yield data that helps people attain the highest possible level of health. Doctors now study health, not disease. Instead of finding out what causes disease, they concentrate on finding out what factors produce the highest possible level of health. The parts of the body that were subject to frequent breakdown, such as the appendix, were eliminated many decades ago by improvements in the genes that blueprint the human body.

Eight months ago the Correlation Center selected Scott and medical scientists from Asia, South America, and Europe to conduct experiments and report on the desirability of increasing the sensitivity of the humans near to 30,000 cycles. Human hearing is limited to a range of about 20 to 20,000 cycles per second. Some animals, such as dogs, can hear higher-pitched sounds. If Scott's committee feels that human well-being would be increased by an addition to the auditory range, research will be conducted toward this end. They already know which DNA manipulation are needed to produce the higher range of hearing. Perhaps in five years a pilot study with 1,000 individuals in different parts of the earth will be tried. If a richer pattern of life results, perhaps 5 per cent of the infants will be so equipped. If results continue to be favorable, this genetic improvement may be generally adopted for future babies of the twenty-second century. If there is any question about the feasibility of a projected improvement, or if difficulties later arise, the flexible scientific attitude results in rapid corrections. Nothing is regarded as final.

During his lifetime Scott knew people who were born in the twentieth century. He considered them "culture-bound." They were so bewildered by the flexibility of the new generations! They kept saying, "It isn't right to do this. If nature had wanted things that way, she would have made them that way. You have no respect for truth." Scot believes that the test of all things is the happiness they yield. He knows that mental straight-jackets limit one's ability to work out patterns of life that are broadly fulfilling.

Perhaps for the first time in human history, people are not bound tightly by the forms of a culture. In the past individuals who did not observe the mores of their particular culture were subjected to penalties that varied from disapproval to death. Scott's generation encourages diversity; they try to avoid getting into personal or social ruts.

In the nurseries children enjoy games that help develop complete flexibility in going from one system to another. They know that two plus two in many situations equals four. They do not want to be rigidly bound by this. They want to know the hidden assumptions that lie behind this ' 'self-evident" formula. Children